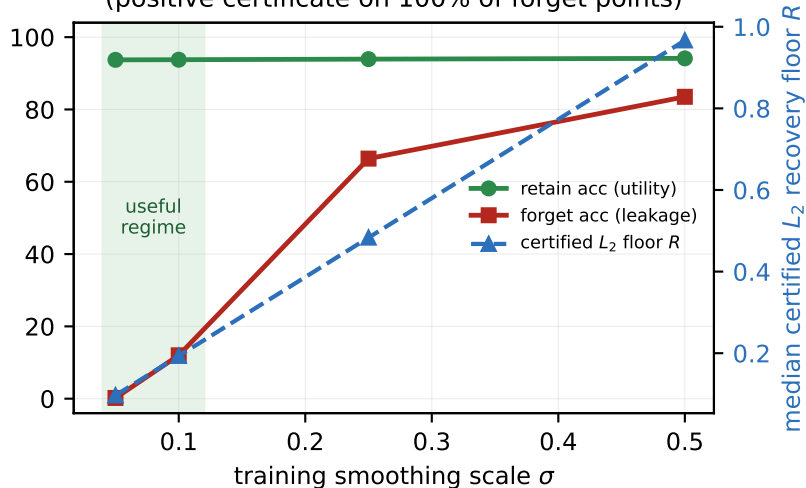


A smoothed unlearning objective with a certified input-space recovery floor (Theorem 4, $\sigma = 0.10$)

(A) Certification--utility frontier
(positive certificate on 100% of forget points)



(B) Same-norm validation: floor respected, retain unrecoverable (% = attack success)

